THE SERENDIB UNIVERSITY ENDOWMENT FUND — EXECUTIVE TEAR SHEET

USD 10 Million Investment Mandate Defense | Quantitative Risk & Extreme Value Theory Engine

Live Cloud App: quant-edge-final-challenge-vnzkxsfkkownmkpp62jboj.streamlit.app • **Code Repository:** github.com/Sahas-sass/quant-edge-final-challenge

OFFICIAL RECOMMENDATION: ACCEPT MANDATE WITH EVT TAIL-RISK BUFFERS

CAPITAL PRESERVATION SCORE: EXCELLENT

The research division recommends accepting the USD 10M endowment mandate under the proposed **Conservative Endowment Allocation**. The portfolio achieves **+7.89% annualized return** (exceeding the university's CPI+4% spending hurdle) with low volatility (**10.40%**), superior drawdown protection (**-28.5%** vs **-34.7%** for 60/40), and rigorous tail defense calibrated via Extreme Value Theory (EVT).

Mandate Size: $10,000,000 USD
Horizon: Perpetual / Long-Term
Regulatory Status: Basel GREEN Zone
EVT 99% 1D ES: 2.75% ($275,160)

1. Recommended Asset Allocation vs Standard Benchmarks

Portfolio	Allocation Weights (%)	Ann. Return	Ann. Volatility	Sharpe (Rf=2%)	Sortino	Max Drawdown	CAPM Beta
Conservative Endowment	AGG:40%, SPY:20%, GLD:15%, EFA:10%, XLE:5%, XLK:5%, EEM:5%	**7.89%**	10.40%	**0.61**	**0.87**	**-28.52%**	0.48
Standard 60/40 Benchmark	SPY: 60%, AGG: 40%	8.55%	11.93%	0.59	0.83	-34.70%	0.60
Equal Weight (1/N)	All 7 assets at 14.3% each	10.24%	16.28%	0.53	0.76	-42.35%	0.77

2. Risk Dashboard: VaR vs Expected Shortfall vs Extreme Value Theory (EVT)

Confidence Level	Parametric VaR	Historical VaR	EVT-POT VaR	Parametric ES	Historical ES	EVT-POT ES	Tail Risk Gap (%)
90% Confidence	0.81%	0.64%	0.64%	1.12%	1.15%	**1.15%**	+3.2%
95% Confidence	1.05%	0.94%	0.93%	1.32%	1.54%	**1.54%**	+17.0%
99% Confidence	1.49%	1.84%	**1.84%**	1.71%	2.76%	**2.75%**	**+60.5%**

3. Crisis Stress Testing on USD 10 Million Portfolio

Crisis Scenario	Market Episode / Narrative	Portfolio Return	Max Drawdown	Dollar Impact ($USD)	Capital Preserved
2008 GFC Lehman Shock	Sep 1, 2008 – Nov 20, 2008 (Global banking panic)	-21.40%	-22.64%	**-$2,139,553**	$7,860,447
March 2020 COVID Crash	Feb 19, 2020 – Mar 23, 2020 (Global liquidity crunch)	-16.98%	-19.07%	**-$1,698,423**	$8,301,577
2022 Fed Rate-Hiking Drawdown	Jan 3, 2022 – Oct 14, 2022 (500bps rate shock)	-15.94%	-16.55%	**-$1,594,042**	$8,405,958
Custom 2026-27 Oil/Rate Shock	Oil +60%, Rates high, Tech -32%, Gold +22%	**-10.75%**	-10.75%	**-$1,075,000**	**$8,925,000**

4. Key Takeaways for the Investment Committee

1. **Superior Risk-Adjusted Profile:** Generates +7.89% return (Sharpe 0.61, Sortino 0.87) vs +8.55% for 60/40, while cutting max drawdown from -34.7% down to -28.5%.
2. **Why EVT is Crucial:** Conventional normal models understate 99% tail loss by **+60.5%**. EVT accurately models fat tails (GPD shape ξ=0.244).
3. **Proven Backtest Reliability:** Passed out-of-sample testing with 16 hits vs 17 expected (Kupiec p=0.8149), placing the fund in the **Basel GREEN Zone**.